

DISEASE PREDICTION AND APPOINTMENT MANAGEMENT SYSTEM

Micro Project Report

Submitted by

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PCCST402 DATABASE MANAGEMENT SYSTEMS

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TABLE OF CONTENTS

S.NO	CONTENTS	PAGE NO.
1	ABSTRACT	3
2	OBJECTIVES OF THE APPLICATION	4
3	ENTITY- RELATIONSHIP DIAGRAM	5
4	DATABASE SCHEMA	6
5	SOFTWARE SPECIFICATION – FRONT END & BACK END	7
6	DATABASE STATE (TABLE DATA)	8
7	DB CONNECTIVITY STEPS/ CODE AND SQL QUERIES	10
8	APPLICATION USER INTERFACE - SCREENSHOTS	13
9	GITHUB LINK	16
10	CONCLUSION	17

ABSTRACT

This project presents the design and development of a **database-driven Healthcare Management Web Application** aimed at improving the efficiency and accuracy of handling healthcare-related information in a digital environment. The system provides a structured platform to **store, retrieve, and manage** core clinical and administrative data such as **patient profiles, doctor and department details, appointments, medical records, prescriptions, laboratory tests and results, and billing/payment transactions**. A Python-based backend is used to implement the business logic and ensure reliable processing of requests, while the frontend built with **HTML and CSS** offers a simple and user-friendly interface for data entry, search, and record viewing.

The application is modeled using an **Entity Relationship (ER) design** to maintain data integrity through well-defined relationships (primary and foreign keys) and to reduce redundancy via normalization. By centralizing records and supporting basic reporting and tracking, the project helps minimize manual paperwork, improves accessibility of patient information, and enables better coordination between clinical, laboratory, and billing modules. Overall, the system demonstrates how a relational database and web technologies can be integrated to create a secure and organized healthcare record management solution suitable for small clinics or as an academic prototype.

OBJECTIVES

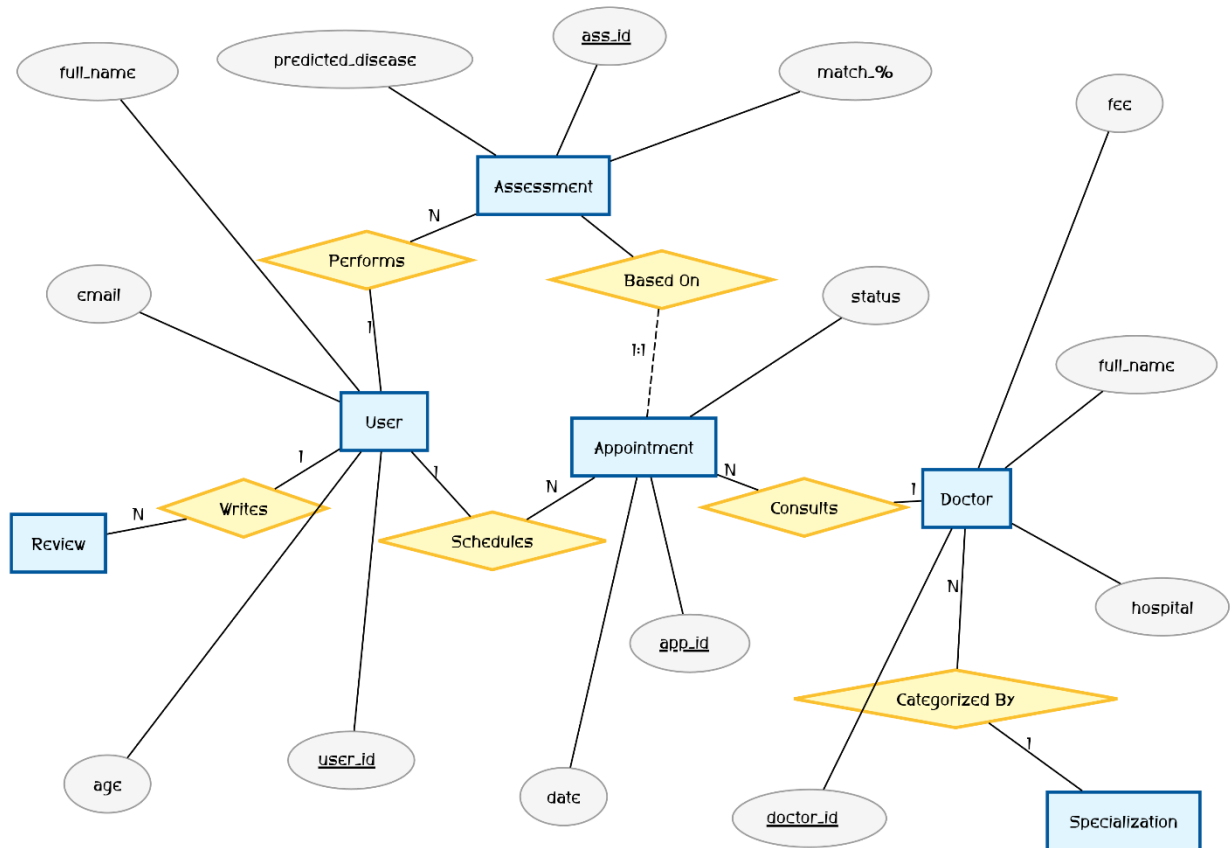
To design an application UI and a normalized database for disease prediction and appointment management system and to perform operations after establishing necessary connectivity between them. Also generate schema diagram and ER diagram for the designed database.

OBJECTIVES:

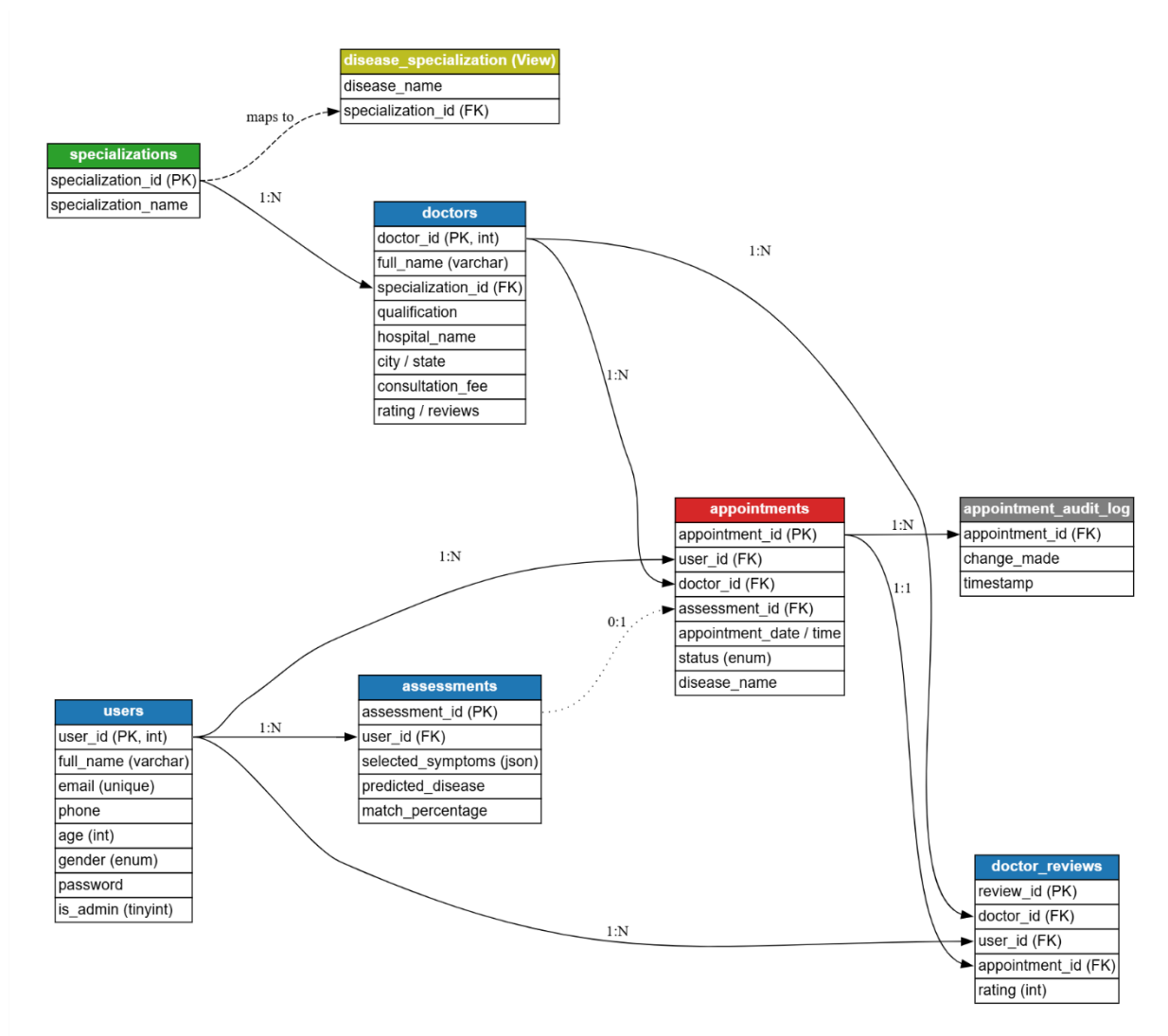
- To develop a database-driven web application that digitally manages healthcare information in a structured and organized manner.
- To store and maintain patient records securely including personal details, visit history, and medical records with proper database relationships (PK/FK).
- To manage doctor and department information to support correct assignment and tracking of consultations.
- To implement appointment scheduling and tracking so patients and doctors can manage visits efficiently (scheduled/completed/cancelled).
- To record diagnoses and prescriptions and link them to the corresponding patient and doctor for accurate clinical history.
- To support laboratory test ordering and result storage to maintain complete patient reports within the system.
- To generate billing invoices and record payments for basic financial tracking and transparency.
- To provide a user-friendly interface (HTML/CSS) for easy data entry, searching, and viewing of healthcare records.
- To ensure data integrity and reduce redundancy through proper ER modeling, normalization, and validation in the backend.

ENTITY-RELATIONSHIP DIAGRAM

ER DIAGRAM FOR THE APPLICATION:



DATABASE SCHEMA



SOFTWARE SPECIFICATION

FRONT END TECHNOLOGIES:

- HTML and CSS - for building and styling the user interface/pages.

BACK END:

- Python - for server-side logic such as handling requests, validations, and database operations.
- Database – MySQL - to store and manage patient, appointment, medical, and billing records.
 - High performance & reliability: MySQL is optimized for fast read/write operations and supports transactions (InnoDB), making it stable and efficient for handling structured records like patients, appointments, and billing.
 - Easy integration & wide support: It integrates smoothly with Python frameworks/libraries (e.g., Flask/Django with MySQL connectors/ORMs) and has broad community, hosting, and tooling support (phpMyAdmin, MySQL Workbench), which is often simpler compared to many alternatives.

DATABASE STATE

• TABLES

```
mysql> use project;
Database changed
mysql> show tables;
+-----+
| Tables_in_project |
+-----+
| appointment_audit_log |
| appointment_overview |
| appointments |
| assessment_audit_log |
| assessments |
| disease_specialization |
| disease_statistics |
| disease_symptoms |
| doctor_reviews |
| doctor_workload |
| doctors |
| high_risk_patients |
| specializations |
| user_health_summary |
| users |
+-----+
15 rows in set (0.02 sec)
```

• USER TABLE

```
mysql> select * from users;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| user_id | full_name | email | phone | age | gender | password | is_admin | created_at | updated_at |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | admin | admin@gmail.com | 9994785654 | 30 | Male | crypt:32768:8:1$UxH09E6Cb8eQJvB8-91d715b91-8e8ee37d2650ff3a291bcc3d48f3c252bd783a0ba6b23dedaf21a59b9a5166e16656e1432f08b3bc5beeb3d48af6afcf153f66b192fbc524f98 | 1 | 2026-02-12 09:48:15 | 2026-02-20 21:12:27 |
| 2 | Admin User | admin@medical.com | 9999999999 | 30 | Other | crypt:32768:8:1$mb7VMW0uVAnhqeHV$7ea2037cf3b381ef8e8d22452b4b6d6b1024a21d58c3c46d2ab37669a1bc82c70870111a0daafe94a700f1fa67cec837b5adfc8322fa283668b4cf47dc48843 | 1 | 2026-02-20 21:12:28 | 2026-02-20 21:33:12 |
| 3 | Pranav N | pn@gmail.com | 88789854875 | 20 | Male | crypt:32768:8:1$LHjt39G72RtMV7qm$33c7b22d562d98238f9969e3263c31bf7ac9f4186f72adaddf2d2b853d4a680b98223f66b3fbd1c24c9e29a01a221bef7f29f44cc9940df69ac7bc8f6857e2a | 0 | 2026-02-20 21:28:16 | 2026-02-20 21:28:16 |
| 4 | sam | sam@gmail.com | 123456789652 | 19 | Male | crypt:32768:8:1$M8IZumKX0MTXs5m$0cfffbe48d8fbc3cfdab232c733886d853af674957d6dade71cd570e1a545dee24d380db216c51d829799a915f22b852f8b70daa43b44ecd7d7afe8603d87909 | 0 | 2026-02-20 22:30:52 | 2026-02-21 11:25:41 |
| 5 | ealiyas shaji | ealiyas@gmail.com | 7854896587 | 25 | Male | crypt:32768:8:1$Uv2E8ZuNb0l4wNgN$c0e3faf9a3d3a8d448f2d910317a085983e720ba15c0ea48e003f2c8343fee6859aa56a6112e32dd05ad06d29168222394b327e569f41e11928aea5da4b1f607 | 0 | 2026-02-20 22:51:20 | 2026-02-20 22:51:20 |
| 6 | ram uday | ramukuttan@gmail.com | 9999954786 | 54 | Female | crypt:32768:8:1$NXEAnWvzq5QD33QU$bf4ce278ed7d4a65871d5e7b9be34f90e8c5bcc0d77c75b9c32b9c09c8aa32cb3cbdc5eda9b16d7d1c8fa5eb2ca132ab7d08abf93e5b1086e847a7ef0e161c | 0 | 2026-02-21 08:49:32 | 2026-02-21 08:49:32 |
| 7 | roshan bejoy | rosh@gmail.com | 212547863 | 10 | Other | crypt:32768:8:1$B0UKZFLYL71M2BnM$c9defafcf84ab393e73198d2695b2abb8fae7f40d282f710be6f6d7d231b5661d5e575311732ac663a38a19ff442520b464b521dad90b1ba37ada2d6e790a62 | 0 | 2026-02-21 11:51:48 | 2026-02-21 11:51:48 |
| 8 | parvathy | paru@gmail.com | 1234567891 | 20 | Female | crypt:32768:8:1$CU7Tvg5366GjtVm$7b75e73bb73fb292bdcc77975d078cba45e6f1aa7ba593bb02278a809486b40f0bb57fc6df6ce9951a5e18134aec493b342cb09aade1c8e58341882d1edf0 | 0 | 2026-02-21 13:10:30 | 2026-02-21 13:10:30 |
| 9 | anju | anjud@gmail.com | 666566552 | 20 | Female | crypt:32768:8:1$MOVJRNlqGfX7GwkS$b15465a9eebbe2b9872de91482f1a170d5b558dd4a08583462d3f2cf69fc71bc22e14673d7fc70c32616e55231db093aa6197116f197d71d8bf8f379a6b50cd9 | 0 | 2026-02-23 12:20:45 | 2026-02-23 12:20:45 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
9 rows in set (0.04 sec)
```

• DOCTORS TABLE

```
mysql> select * from doctors;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| doctor_id | full_name | specialization_id | qualification | experience_years | hospital_name | city | state |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Dr. Suresh Menon | 1 | MBBS, MD (Neurology) | 6 | Max Hospital | Mumbai | Maharashtra | | |
| 106 | MG Road, Mumbai | +91-9606008142 | NULL | 1027.00 | 3.76 | 96 | Available | NULL | 2026-02-12 11:46:22 |
| 2 | Dr. Kavita Desai | 6 | MBBS, DNB (Medicine) | 8 | Manipal Hospital | Mumbai | Maharashtra |
| 210 | MG Road, Mumbai | +91-8499402700 | NULL | 819.00 | 3.95 | 245 | Available | NULL | 2026-02-12 11:46:22 |
| 3 | Dr. Sachin Naidu | 1 | MBBS, MD (Internal Medicine) | 22 | Continental Hospital | Mumbai | Maharashtra |
| 92 | MG Road, Mumbai | +91-9191010756 | NULL | 1720.00 | 4.61 | 93 | Busy | NULL | 2026-02-12 11:46:22 |
| 4 | Dr. Arun Verma | 6 | MBBS, MD (Medicine) | 16 | Lilavati Hospital | Delhi | Delhi |
| 553 | MG Road, Delhi | +91-7138845907 | NULL | 1304.00 | 4.25 | 355 | Busy | NULL | 2026-02-12 11:46:22 |
| 5 | Dr. Amit Patel | 1 | MBBS, MD (Medicine) | 23 | BLK Hospital | Delhi | Delhi |
| 773 | MG Road, Delhi | +91-9167266283 | NULL | 1475.00 | 3.78 | 327 | Available | NULL | 2026-02-12 11:46:22 |
| 6 | Dr. Nidhi Malhotra | 1 | MBBS, DNB (Medicine) | 10 | Columbia Asia | Delhi | Delhi |
| 792 | MG Road, Delhi | +91-8239985943 | NULL | 1088.00 | 4.74 | 208 | Busy | NULL | 2026-02-12 11:46:22 |
| 7 | Dr. Vidya Bhatt | 1 | MBBS, MD, DM (Neurology) | 19 | Columbia Asia | Bangalore | Karnataka |
| 574 | MG Road, Bangalore | +91-8846525389 | NULL | 1262.00 | 4.49 | 332 | Available | NULL | 2026-02-12 11:46:22 |
| 8 | Dr. Sunita Rao | 1 | MBBS, MD (Medicine) | 22 | Global Hospital | Bangalore | Karnataka |
| 691 | MG Road, Bangalore | +91-7457421426 | NULL | 1582.00 | 4.29 | 486 | Busy | NULL | 2026-02-12 11:46:22 |
| 9 | Dr. Suresh Menon | 1 | MBBS, MD (Neurology) | 9 | BLK Hospital | Bangalore | Karnataka |
| 781 | MG Road, Bangalore | +91-8855535876 | NULL | 1537.00 | 3.66 | 412 | Available | NULL | 2026-02-12 11:46:22 |
| 10 | Dr. Priya Sharma | 3 | MBBS, MD (Internal Medicine) | 14 | Max Hospital | Bangalore | Karnataka |
| 68 | MG Road, Bangalore | +91-8825947819 | NULL | 1675.00 | 3.84 | 354 | Available | NULL | 2026-02-12 11:46:22 |
| 11 | Dr. Priyanka Das | 8 | MBBS, MD (Internal Medicine) | 10 | Continental Hospital | Bangalore | Karnataka |
| 616 | MG Road, Bangalore | +91-79064085216 | NULL | 1912.00 | 4.44 | 287 | Available | NULL | 2026-02-12 11:46:22 |
| 12 | Dr. Rekha Pandey | 1 | MBBS, DNB (Medicine) | 25 | Max Hospital | Hyderabad | Telangana |
| 156 | MG Road, Hyderabad | +91-8051831476 | NULL | 807.00 | 3.55 | 127 | Busy | NULL | 2026-02-12 11:46:22 |
| 13 | Dr. Shruti Kapoor | 1 | MBBS, MD (Neurology) | 10 | Continental Hospital | Hyderabad | Telangana |
| 289 | MG Road, Hyderabad | +91-7800974465 | NULL | 1012.00 | 4.09 | 67 | Available | NULL | 2026-02-12 11:46:22 |
| 14 | Dr. Priyanka Das | 2 | MBBS, MD (Internal Medicine) | 15 | BLK Hospital | Hyderabad | Telangana |
| 740 | MG Road, Hyderabad | +91-7983702588 | NULL | 1201.00 | 4.22 | 232 | Available | NULL | 2026-02-12 11:46:22 |
```


• APPOINTMENT TABLE

```
mysql> SELECT * FROM APPOINTMENTS;
```

appointment_id	user_id	doctor_id	assessment_id	appointment_date	appointment_time	status	disease_name	symptoms
1	1	3	3	2026-02-19	16:00:00	Pending	Paralysis (brain hemorrhage)	["joint_pain", "cough"]
2	1	21	21	2026-02-12 11:54:29	2026-02-12 11:54:29	Pending	Paralysis (brain hemorrhage)	["fever", "nausea", "skin_rash", "cough"]
3	1	3	3	2026-02-18	12:00:00	Pending	Paralysis (brain hemorrhage)	["weight_loss", "yellow_eyes"]
4	1	49	49	2026-02-12 12:08:55	2026-02-12 12:08:55	Pending	Paralysis (brain hemorrhage)	["weight_loss", "yellow_eyes"]
5	1	1	1	2026-02-19	11:00:00	Pending	Paralysis (brain hemorrhage)	["joint_pain", "skin_rash"]
7	4	48	48	2026-02-12 12:16:23	2026-02-12 12:16:23	Pending	Paralysis (brain hemorrhage)	["headache", "nausea", "cough", "yellow_eyes"]
8	4	1	1	2026-02-21 08:48:42	2026-02-21 08:48:42	Pending	Paralysis (brain hemorrhage)	["fever"]
9	7	21	21	2026-02-21 11:24:37	2026-02-21 11:24:37	Pending	Paralysis (brain hemorrhage)	["headache"]
11	4	49	49	2026-02-21 11:55:51	2026-02-21 11:55:51	Pending	Paralysis (brain hemorrhage)	["fever", "nausea", "fatigue", "cough", "weight_loss"]
255			255	2026-02-23 12:17:09	2026-02-23 12:17:09			

9 rows in set (0.02 sec)

• HIGH RISH PATIENTS TABLE

```
mysql> select * from high_risk_patients;
```

user_id	full_name	email	age	gender	assessment_id	predicted_disease	match_percentage	assessment_date
4	sam	sam@gmail.com	19	Male	52	Paralysis (brain hemorrhage)	100.00	2026-02-23 12:16:13
4	sam	sam@gmail.com	19	Male	51	Cervical spondylosis	100.00	2026-02-23 12:14:53
4	sam	sam@gmail.com	19	Male	48	Impetigo	100.00	2026-02-23 12:13:52
4	sam	sam@gmail.com	19	Male	47	Hepatitis B	100.00	2026-02-23 12:13:30
4	sam	sam@gmail.com	19	Male	46	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:19:26
4	sam	sam@gmail.com	19	Male	44	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:18:28
4	sam	sam@gmail.com	19	Male	43	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:18:13
4	sam	sam@gmail.com	19	Male	42	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:17:53
8	parvathy	paru@gmail.com	20	Female	41	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:15:20
8	parvathy	paru@gmail.com	20	Female	40	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:15:08
8	parvathy	paru@gmail.com	20	Female	39	Paralysis (brain hemorrhage)	100.00	2026-02-21 13:13:07
8	parvathy	paru@gmail.com	20	Female	36	Osteoarthritis	100.00	2026-02-21 13:12:14
8	parvathy	paru@gmail.com	20	Female	35	Hypertension	100.00	2026-02-21 13:11:19
7	roshan bejoy	rosh@gmail.com	10	Other	34	Paralysis (brain hemorrhage)	100.00	2026-02-21 11:55:19
7	roshan bejoy	rosh@gmail.com	10	Other	33	Tuberculosis	100.00	2026-02-21 11:53:58
7	roshan bejoy	rosh@gmail.com	10	Other	32	Paralysis (brain hemorrhage)	100.00	2026-02-21 11:53:52
4	sam	sam@gmail.com	19	Male	31	Hypertension	100.00	2026-02-21 11:50:43
4	sam	sam@gmail.com	19	Male	30	Paralysis (brain hemorrhage)	100.00	2026-02-21 11:50:34
4	sam	sam@gmail.com	19	Male	29	Paralysis (brain hemorrhage)	100.00	2026-02-21 11:23:54
4	sam	sam@gmail.com	19	Male	28	Paralysis (brain hemorrhage)	100.00	2026-02-21 11:23:43
6	ram uday	ramukuttan@gmail.com	54	Female	27	Hypertension	100.00	2026-02-21 11:23:04
4	sam	sam@gmail.com	19	Male	26	Paralysis (brain hemorrhage)	100.00	2026-02-21 08:50:33
5	ealiyas shaji	ealiyas@gmail.com	25	Male	25	Paralysis (brain hemorrhage)	100.00	2026-02-21 08:47:59
5	ealiyas shaji	ealiyas@gmail.com	25	Male	24	Paralysis (brain hemorrhage)	100.00	2026-02-20 22:55:13
5	ealiyas shaji	ealiyas@gmail.com	25	Male	23	Paralysis (brain hemorrhage)	100.00	2026-02-20 22:54:26
5	ealiyas shaji	ealiyas@gmail.com	25	Male	22	Drug Reaction	100.00	2026-02-20 22:54:11
3	Pranav N	pn@gmail.com	20	Male	20	Paralysis (brain hemorrhage)	100.00	2026-02-20 22:22:03
1	admin	admin@gmail.com	30	Male	19	Hypothyroidism	100.00	2026-02-20 20:55:38

• SPECIALIZATIONS

```
mysql> select * from specializations
```

```

-> ;

```

specialization_id	specialization_name	description	created_at
1	Neurologist	Specializes in disorders of the nervous system including brain hemorrhage	2026-02-12 11:46:21
2	General Physician	Treats general health conditions and provides primary care	2026-02-12 11:46:21
3	Internal Medicine	Focuses on adult diseases and complex diagnostic problems	2026-02-12 11:46:21
4	Emergency Medicine	Handles acute and urgent medical conditions	2026-02-12 11:46:21
5	Cardiologist	Specializes in heart and cardiovascular conditions	2026-02-12 11:46:21
6	Pulmonologist	Treats respiratory system and lung diseases	2026-02-12 11:46:21
7	Gastroenterologist	Specializes in digestive system disorders	2026-02-12 11:46:21
8	Infectious Disease Specialist	Treats infectious diseases and fever-related conditions	2026-02-12 11:46:21

8 rows in set (0.01 sec)

DB CONNECTIVITY STEPS/ CODE & SQL QUERIES

1) MySQL DB Connectivity Steps.

- i) Install MySQL Server + MySQL Workbench
- ii) Create Database
- iii) Create Tables
- iv) Install Python MySQL library
- v) Configure DB credentials in Python
- vi) Test connection
- vii) Perform CRUD operations

2) Python Connectivity Code

```
from flask import Flask, request, jsonify
import mysql.connector
```

```
app = Flask(__name__)
```

```
def get_connection():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="YOUR_PASSWORD",
        database="healthcare_db"
    )
```

```
@app.route("/patients", methods=["POST"])
```

```
def add_patient():
```

```
    data = request.json
    conn = get_connection()
    cursor = conn.cursor()
```

```
    sql = """
        INSERT INTO patient (first_name, last_name, dob, gender, phone, email, address)
        VALUES (%s, %s, %s, %s, %s, %s, %s)
    """
```

```
    values = (
        data["first_name"], data["last_name"], data["dob"], data["gender"],
        data["phone"], data["email"], data["address"]
    )
```

```
    cursor.execute(sql, values)
```

```
conn.commit()

new_id = cursor.lastrowid
cursor.close()
conn.close()

return jsonify({"message": "Patient added", "patient_id": new_id}), 201

@app.route("/patients/<int:patient_id>", methods=["GET"])
def get_patient(patient_id):
    conn = get_connection()
    cursor = conn.cursor(dictionary=True)

    cursor.execute("SELECT * FROM patient WHERE patient_id=%s", (patient_id,))
    row = cursor.fetchone()

    cursor.close()
    conn.close()

    if not row:
        return jsonify({"message": "Patient not found"}), 404

    return jsonify(row), 200

if __name__ == "__main__":
    app.run(debug=True)
```

3) INSERTION

```
INSERT INTO patient (first_name, last_name, dob, gender, phone, email, address)
VALUES ('Amit', 'Sharma', '2002-04-12', 'Male', '9876543210', 'amit@gmail.com', 'Pune');

INSERT INTO doctor (first_name, last_name, specialization, phone, email)
VALUES ('Neha', 'Patel', 'Cardiology', '9123456780', 'neha@clinic.com');

INSERT INTO appointment (patient_id, doctor_id, appointment_date, appointment_time,
reason)
VALUES (1, 1, '2026-04-01', '10:30:00', 'Routine checkup');
```

4) SELECT

```
SELECT * FROM patient;
```

```
SELECT * FROM appointment WHERE appointment_date = '2026-04-01';
```

```
SELECT
  a.appointment_id, a.appointment_date, a.appointment_time, a.status,
  p.first_name AS patient_first, p.last_name AS patient_last,
  d.first_name AS doctor_first, d.last_name AS doctor_last
FROM appointment a
JOIN patient p ON a.patient_id = p.patient_id
JOIN doctor d ON a.doctor_id = d.doctor_id;
```

5) DELETION

```
DELETE FROM appointment WHERE appointment_id = 1;
DELETE FROM patient WHERE patient_id = 1;
```

6) View

```
USE healthcare_db;
```

```
CREATE OR REPLACE VIEW vw_patient_appointment_summary AS
SELECT
  p.patient_id,
  CONCAT(p.first_name, ' ', p.last_name) AS patient_name,
  COUNT(a.appointment_id) AS total_appointments
FROM patient p
LEFT JOIN appointment a ON a.patient_id = p.patient_id
GROUP BY p.patient_id, p.first_name, p.last_name;
```

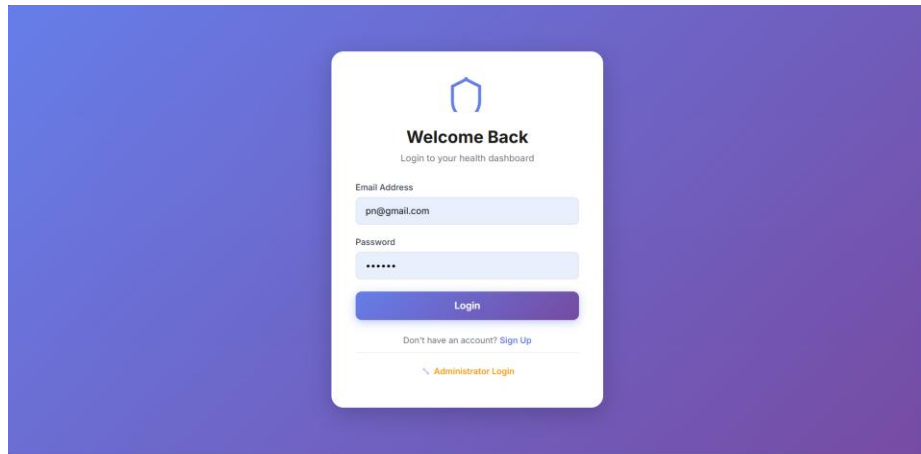
7) TRIGGER

```
DELIMITER $$
CREATE TRIGGER trg_patient_phone_update
AFTER UPDATE ON patient
FOR EACH ROW
BEGIN
  IF OLD.phone <> NEW.phone THEN
    INSERT INTO patient_audit (patient_id, action_type, old_phone, new_phone)
    VALUES (OLD.patient_id, 'UPDATE', OLD.phone, NEW.phone);
  END IF;
END $$
```

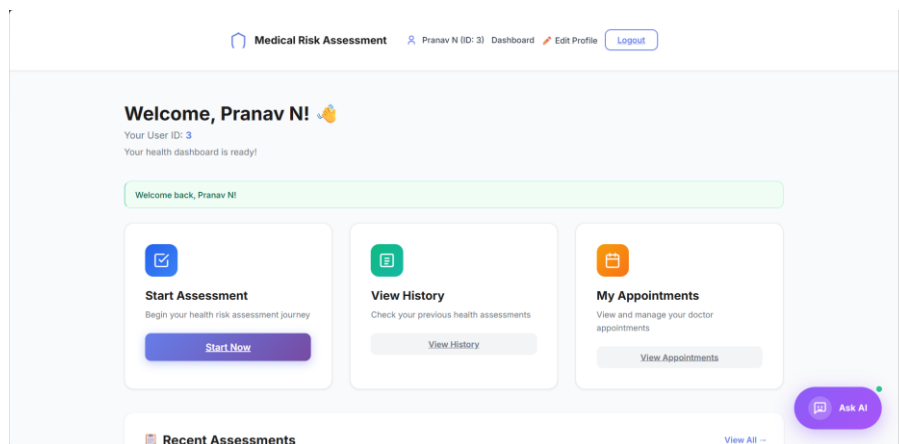
```
DELIMITER ;
```

APPLICATION USER INTERFACE

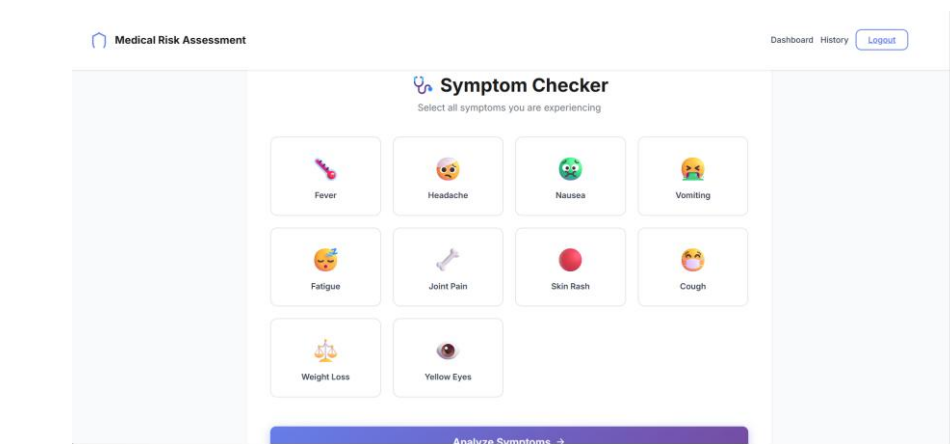
- **LOGIN PAGE (USER)**



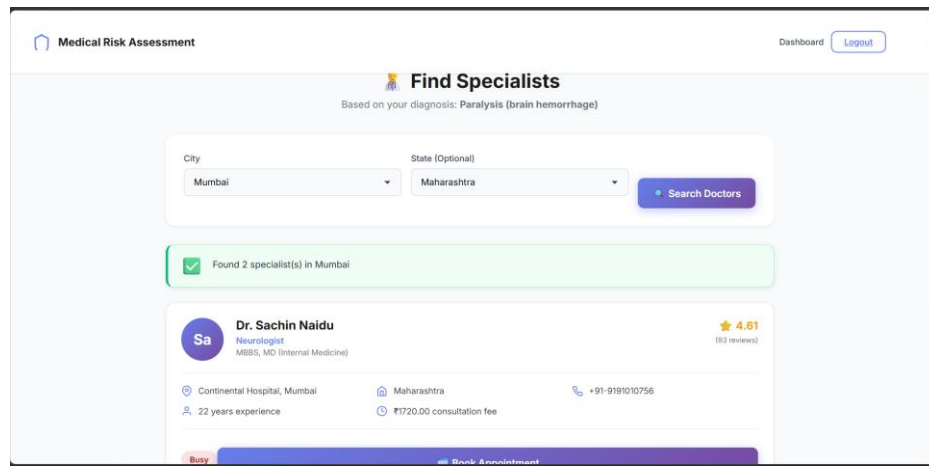
- **HOME PAGE**



- **SYMPTOM CHECKING PAGE**

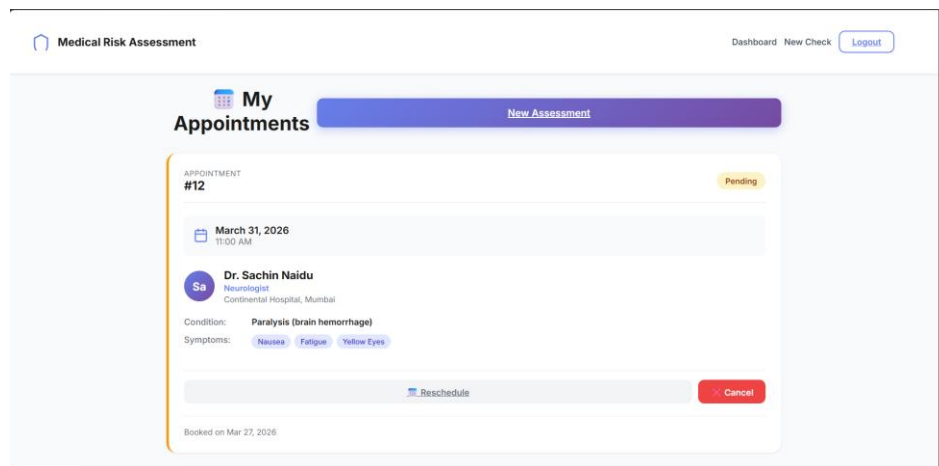


- **APPOINTMENT BOOKING PAGE**



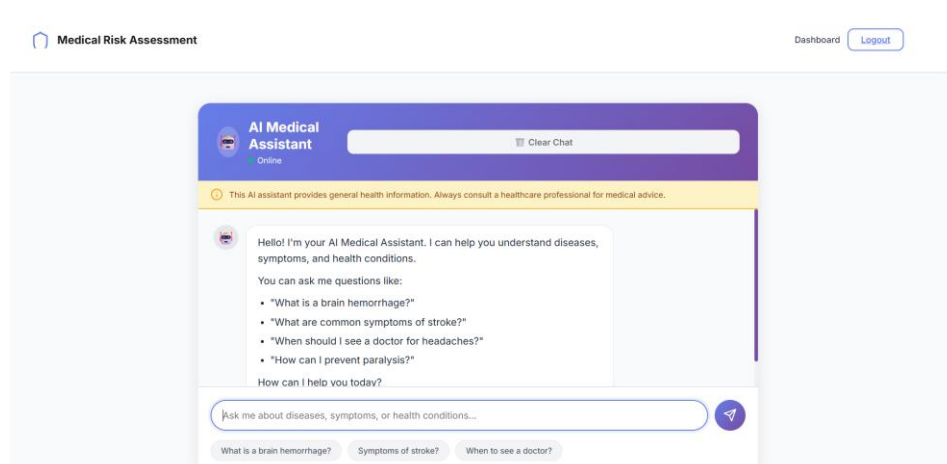
The screenshot shows the 'Find Specialists' section of the Medical Risk Assessment dashboard. The user's diagnosis is 'Paralysis (brain hemorrhage)'. The search filters are set to City: Mumbai and State: Maharashtra. A 'Search Doctors' button is visible. Below the search bar, a green notification bar states 'Found 2 specialist(s) in Mumbai'. The first specialist listed is Dr. Sachin Naidu, a Neurologist with a 4.61 rating (93 reviews). His profile includes his hospital (Continental Hospital, Mumbai), location (Maharashtra), phone number (+91-9191010756), and 22 years of experience. A 'Book Appointment' button is at the bottom of the specialist's profile.

- **BOOKING DETAIL PAGE**



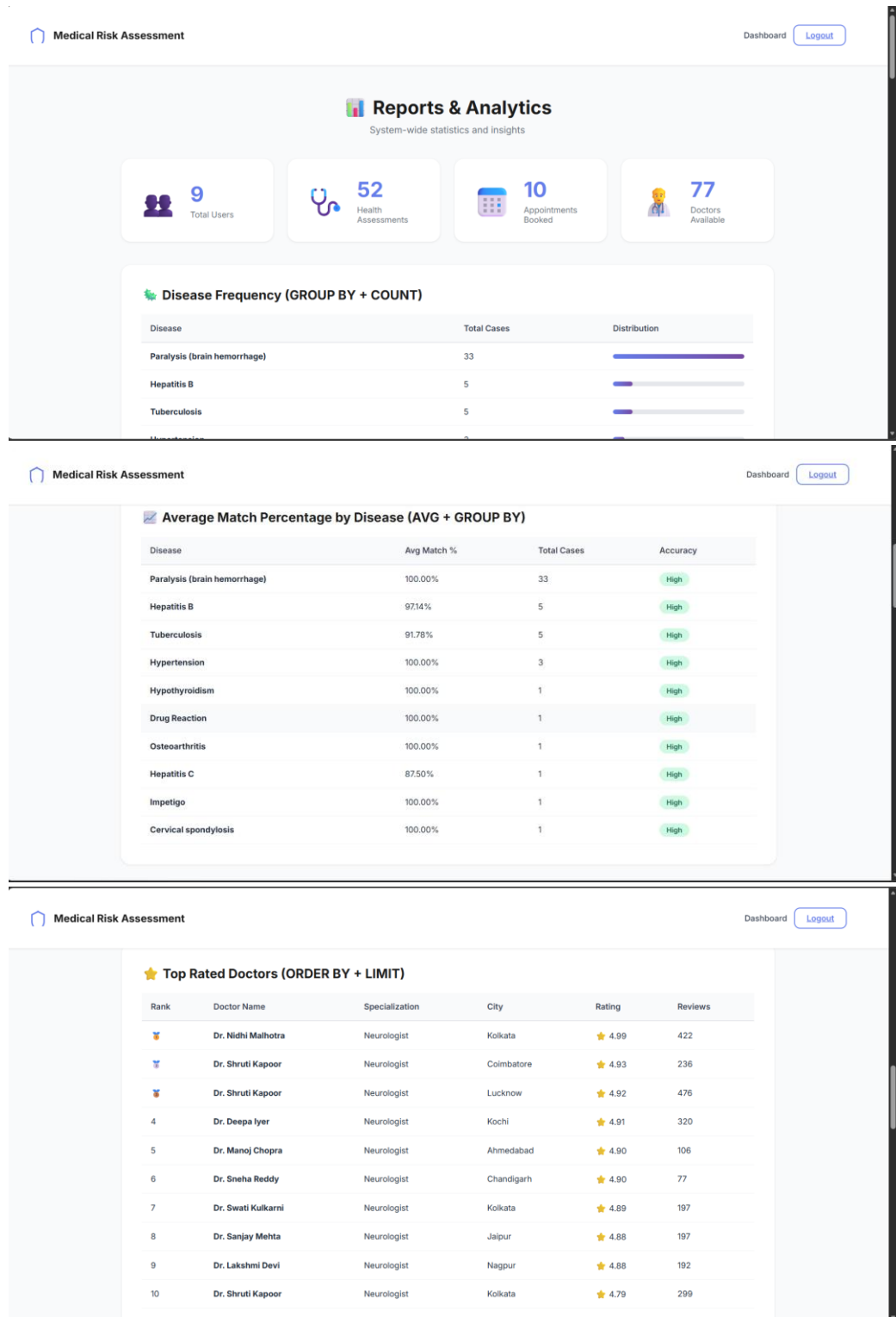
The screenshot shows the 'My Appointments' section of the Medical Risk Assessment dashboard. A 'New Assessment' button is at the top right. The appointment details for 'APPOINTMENT #12' are shown, with a status of 'Pending'. The appointment is scheduled for March 31, 2026, at 11:00 AM. The specialist is Dr. Sachin Naidu, a Neurologist at Continental Hospital, Mumbai. The condition is 'Paralysis (brain hemorrhage)' and the symptoms are 'Nausea', 'Fatigue', and 'Yellow Eyes'. There are 'Reschedule' and 'Cancel' buttons at the bottom. The appointment was booked on Mar 27, 2026.

- **MEDICAL ASSISTANT PAGE**



The screenshot shows the 'AI Medical Assistant' chat interface. The assistant is online and provides general health information. A disclaimer states: 'This AI assistant provides general health information. Always consult a healthcare professional for medical advice.' The assistant's message reads: 'Hello! I'm your AI Medical Assistant. I can help you understand diseases, symptoms, and health conditions. You can ask me questions like: • "What is a brain hemorrhage?" • "What are common symptoms of stroke?" • "When should I see a doctor for headaches?" • "How can I prevent paralysis?" How can I help you today?'. There is a text input field for the user to ask questions, with a 'Send' button. Below the input field are three suggested questions: 'What is a brain hemorrhage?', 'Symptoms of stroke?', and 'When to see a doctor?'.

• ADMIN HOME PAGE



GITHUB LINK

GITHUB LINK: <https://github.com/Pranav-syntax/healthcare>

CONCLUSION

The proposed database-driven Healthcare Management Web Application successfully demonstrates how clinical and administrative activities can be digitized using a structured MySQL database with a Python backend and a web-based interface. The system centralizes important information such as patient details, doctor data, appointments, and related records, making storage and retrieval faster and more organized compared to manual paperwork. The overall design focuses on practical usability, and the UI was intentionally kept simple and easy to navigate so that elderly users can use it comfortably without confusion.

One of the main advantages of this project is efficient and reliable data management—MySQL ensures data integrity through relationships (primary/foreign keys) and supports quick searching and updating of records. In addition, the system reduces redundancy, improves accuracy, and saves time for staff by automating common tasks like record lookup and appointment tracking. With a clean interface and database-backed structure, the project offers a scalable foundation that can be extended in future with modules such as advanced reporting, role-based login, and enhanced security features.

Marks Awarded:	_____/15
Micro Project Verified by	
Signature of Course Faculty:	
Name of Course Faculty:	